

Operating Systems Course

Project Cover Sheets

Algorithm Comparison Projects

Student-facing cover sheets and submission guide for the five approved Part 1 comparison projects.

Included in this document:

- one cover sheet template per project
- one separate variant-specific checklist page for each project
- detailed submission requirements using **any programming language**
- GitHub and Project Submission Form guidance
- final submission checklist with evaluator notes

Projects

1. Priority vs SRTF Comparison Project
2. Round Robin vs SRTF Comparison Project
3. Round Robin vs Priority Comparison Project
4. SJF vs Priority Comparison Project
5. Round Robin vs SJF Comparison Project

Priority vs SRTF Comparison Project

Cover Sheet - Algorithm Comparison Project

Project Title	Priority vs SRTF Comparison Project	Team Number	
Repository URL			

Team Members

No.	Student Name	Student ID	Contribution Area	Grade	Notes
1					
2					
3					
4					
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Evaluation Rubric Summary

Criterion	Marks	Specific focus	Notes for evaluator
Interface and usability	15	-	Clean layout, easy data entry, easy comparison workflow, readable output.
Input handling and validation	10	-	Reject invalid numeric input, duplicate IDs, missing values, invalid quantum/priority values.
Algorithm A correctness	15	Priority Scheduling	Correct process selection, arrival handling, ties, completion behavior, and algorithm-specific rules.
Algorithm B correctness	15	SRTF	Correct process selection, arrival handling, ties, completion behavior, and algorithm-specific rules.
Gantt charts and execution	10	-	Separate readable Gantt

visualization			chart for each algorithm with correct time markers and execution order.
Metrics calculation	10	-	Correct WT, TAT, RT for each process and correct averages, consistent with charts.
Comparison quality and fairness	10	Both algorithms	Same workload, correct comparison tables, valid analysis of trade-offs and recommendation.
Required test scenarios	5	-	Normal case, behavior-revealing case, and invalid-input validation case are all included.
Report and technical explanation	5	-	Clear write-up, screenshots, assumptions, limitations, and technically correct conclusions.
Viva / team understanding	5	-	Team members can explain both algorithms, outputs, metrics, and design decisions.

Variant-Specific Checklist

Use this page during review. A project should not be considered complete until all variant-specific items have been checked.

Check	Checklist Item	Notes
☐	Priority rule is stated clearly and applied consistently.	
☐	Ties in priority are resolved using a documented rule.	
☐	SRTF preemption points are correct and visible in the Gantt chart.	
☐	At least one workload shows conflict between priority and burst time.	

□	Fairness and starvation are discussed in the conclusion.	
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Project-Specific Notes for Students

- Use the same workload in both algorithms.
- Clearly state the priority rule: smaller value = higher priority, or the opposite.
- Assume **preemptive** Priority Scheduling.
- Show and explain **preemption** in SRTF whenever a shorter remaining job arrives.
- Discuss fairness, starvation risk, and differences between policy-based service and burst-time-based service.

Round Robin vs SRTF Comparison Project

Cover Sheet - Algorithm Comparison Project

Project Title	Round Robin vs SRTF Comparison Project	Team Number	
Repository URL			

Team Members

No.	Student Name	Student ID	Contribution Area	Grade	Notes
1					
2					
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Evaluation Rubric Summary

Criterion	Marks	Specific focus	Notes for evaluator
Interface and usability	15	-	Clean layout, easy data entry, easy comparison workflow, readable output.
Input handling and validation	10	-	Reject invalid numeric input, duplicate IDs, missing values, invalid quantum/priority values.
Algorithm A correctness	15	Round Robin	Correct process selection, arrival handling, ties, completion behavior, and algorithm-specific rules.
Algorithm B correctness	15	SRTF	Correct process selection, arrival handling, ties, completion behavior, and algorithm-specific rules.

Gantt charts and execution visualization	10	-	Separate readable Gantt chart for each algorithm with correct time markers and execution order.
Metrics calculation	10	-	Correct WT, TAT, RT for each process and correct averages, consistent with charts.
Comparison quality and fairness	10	Both algorithms	Same workload, correct comparison tables, valid analysis of trade-offs and recommendation.
Required test scenarios	5	-	Normal case, behavior-revealing case, and invalid-input validation case are all included.
Report and technical explanation	5	-	Clear write-up, screenshots, assumptions, limitations, and technically correct conclusions.
Viva / team understanding	5	-	Team members can explain both algorithms, outputs, metrics, and design decisions.

Variant-Specific Checklist

Use this page during review. A project should not be considered complete until all variant-specific items have been checked.

Check	Checklist Item	Notes
☐	Time quantum is stated clearly in the interface and report.	
☐	Round Robin ready queue behavior is correct.	
☐	SRTF preemption behavior is correct.	
☐	At least one workload reveals a fairness difference between the two algorithms.	

☐	Fairness versus efficiency discussed	
☐	The conclusion explains the effect of the chosen quantum.	

Project-Specific Notes for Students

- Use the same workload in both algorithms.
- State the selected time quantum clearly and justify it.
- Round Robin queue rotation and arrivals during execution must be handled correctly.
- Discuss fairness versus efficiency and the effect of time slicing on response time.

Round Robin vs Priority Comparison Project

Cover Sheet – Algorithm Comparison Project

Project Title	Round Robin vs Priority Comparison Project	Team Number	
Repository URL			

Team Members

No.	Student Name	Student ID	Contribution Area	Grade	Notes
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2					
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Evaluation Rubric Summary

Criterion	Marks	Specific focus	Notes for evaluator
Interface and usability	15	-	Clean layout, easy data entry, easy comparison workflow, readable output.
Input handling and validation	10	-	Reject invalid numeric input, duplicate IDs, missing values, invalid quantum/priority values.
Algorithm A correctness	15	Round Robin	Correct process selection, arrival handling, ties, completion behavior, and algorithm-specific rules.
Algorithm B correctness	15	Priority Scheduling	Correct process selection, arrival handling, ties, completion behavior, and algorithm-specific rules.

Gantt charts and execution visualization	10	-	Separate readable Gantt chart for each algorithm with correct time markers and execution order.
Metrics calculation	10	-	Correct WT, TAT, RT for each process and correct averages, consistent with charts.
Comparison quality and fairness	10	Both algorithms	Same workload, correct comparison tables, valid analysis of trade-offs and recommendation.
Required test scenarios	5	-	Normal case, behavior-revealing case, and invalid-input validation case are all included.
Report and technical explanation	5	-	Clear write-up, screenshots, assumptions, limitations, and technically correct conclusions.
Viva / team understanding	5	-	Team members can explain both algorithms, outputs, metrics, and design decisions.

Variant-Specific Checklist

Use this page during review. A project should not be considered complete until all variant-specific items have been checked.

Check	Checklist Item	Notes
☐	Quantum value is stated clearly.	
☐	Priority rule is stated clearly and applied consistently.	
☐	Round Robin queue behavior is correct.	
☐	At least one urgency-focused workload is included.	

☐	Starvation or unfair delay is discussed where relevant.	
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Project-Specific Notes for Students

- Use the same workload in both algorithms.
- State both the quantum value and the priority rule clearly.
- Assume **preemptive** Priority Scheduling.
- Discuss fairness, urgency, service differentiation, and starvation risk.
- Include a workload where high-priority jobs are expected to gain an advantage.

SJF vs Priority Comparison Project

Cover Sheet - Algorithm Comparison Project

Project Title	SJF vs Priority Comparison Project	Team Number	
Repository URL			

Team Members

No.	Student Name	Student ID	Contribution Area	Grade	Notes
1					
2					
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Evaluation Rubric Summary

Criterion	Marks	Specific focus	Notes for evaluator
Interface and usability	15	-	Clean layout, easy data entry, easy comparison workflow, readable output.
Input handling and validation	10	-	Reject invalid numeric input, duplicate IDs, missing values, invalid quantum/priority values.
Algorithm A correctness	15	SJF	Correct process selection, arrival handling, ties, completion behavior, and algorithm-specific rules.
Algorithm B correctness	15	Priority Scheduling	Correct process selection, arrival handling, ties, completion behavior, and algorithm-specific rules.

Gantt charts and execution visualization	10	-	Separate readable Gantt chart for each algorithm with correct time markers and execution order.
Metrics calculation	10	-	Correct WT, TAT, RT for each process and correct averages, consistent with charts.
Comparison quality and fairness	10	Both algorithms	Same workload, correct comparison tables, valid analysis of trade-offs and recommendation.
Required test scenarios	5	-	Normal case, behavior-revealing case, and invalid-input validation case are all included.
Report and technical explanation	5	-	Clear write-up, screenshots, assumptions, limitations, and technically correct conclusions.
Viva / team understanding	5	-	Team members can explain both algorithms, outputs, metrics, and design decisions.

Variant-Specific Checklist

Use this page during review. A project should not be considered complete until all variant-specific items have been checked.

Check	Checklist Item	Notes
☐	SJF selection is based on the shortest available burst time.	
☐	Priority rule is stated clearly and applied consistently.	
☐	At least one workload shows conflict between short burst time and high priority.	
☐	Comparison tables use the same dataset for both	

	algorithms.	
□	Conclusion explains efficiency versus urgency trade-off.	

Project-Specific Notes for Students

- Use the same workload in both algorithms.
- Assume **preemptive** SJF and Priority Scheduling.
- Discuss efficiency versus urgency when short jobs and high-priority jobs conflict.
- Explain whether either policy causes unfair delay.

Round Robin vs SJF Comparison Project

Cover Sheet - Algorithm Comparison Project

Project Title	Round Robin vs SJF Comparison Project	Team Number	
Repository URL			

Team Members

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Evaluation Rubric Summary

Criterion	Marks	Specific focus	Notes for evaluator
Interface and usability	15	-	Clean layout, easy data entry, easy comparison workflow, readable output.
Input handling and validation	10	-	Reject invalid numeric input, duplicate IDs, missing values, invalid quantum/priority values.
Algorithm A correctness	15	Round Robin	Correct process selection, arrival handling, ties, completion behavior, and algorithm-specific rules.
Algorithm B correctness	15	SJF	Correct process selection, arrival handling, ties, completion behavior, and algorithm-specific rules.

Gantt charts and execution visualization	10	-	Separate readable Gantt chart for each algorithm with correct time markers and execution order.
Metrics calculation	10	-	Correct WT, TAT, RT for each process and correct averages, consistent with charts.
Comparison quality and fairness	10	Both algorithms	Same workload, correct comparison tables, valid analysis of trade-offs and recommendation.
Required test scenarios	5	-	Normal case, behavior-revealing case, and invalid-input validation case are all included.
Report and technical explanation	5	-	Clear write-up, screenshots, assumptions, limitations, and technically correct conclusions.
Viva / team understanding	5	-	Team members can explain both algorithms, outputs, metrics, and design decisions.

Variant-Specific Checklist

Use this page during review. A project should not be considered complete until all variant-specific items have been checked.

Check	Checklist Item	Notes
☐	Quantum value is stated clearly.	
☐	Round Robin queue behavior is correct.	
☐	SJF chooses the shortest available burst when the CPU becomes free.	
☐	At least one workload reveals long-job sensitivity or fairness difference.	

□	Conclusion explains fairness versus efficiency trade-off.	
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Project-Specific Notes for Students

- Use the same workload in both algorithms.
- Assume **preemptive** SJF.
- State the Round Robin quantum clearly.
- Discuss fairness versus efficiency, especially for long jobs and first response time.

Detailed Submission Description

The following requirements apply to all comparison projects.

Implementation Technology

- Programming language: **any programming language**.
- GUI technology: must be used for the interface and visualization components.

GitHub Submission Guidelines

- Repository: the full project must be submitted through GitHub in a clean repository with meaningful commits.
- The repository must include source code, README, screenshots, sample test cases, and run instructions.
- The repository should include a clear package structure and separate classes for models, scheduling logic, GUI, metrics, and utilities.

Project Submission Form Guidelines

- A Project Submission Form must be completed and submitted with the GitHub repository link, project title, team number, and student information.
- The default branch must build and run correctly at submission time.

Recommended Repository Structure

```
project-root/
  src/
    model/
    scheduler/
    metrics/
    gui/
    util/
  assets/
  screenshots/
  test-cases/
  README.md
  .gitignore
```

Submission Checklist with Notes

Done	Submission Item	Notes
☐	GitHub repository link provided	The submitted link must be accessible and point to the final repository.
☐	README included	Must contain project description, requirements, build/run steps, and team member names.
☐	Source code included	Do not submit compiled files only. Source code must be complete.
☐	Executable/runnable	List Java version, JavaFX setup, IDE or

	instructions included	command-line execution steps.
☐	Screenshots included	Include interface screenshots and at least one Gantt chart screenshot per algorithm.
☐	Test scenarios included	Provide at least three documented test scenarios with outputs.
☐	Project Submission Form completed	Must match the repository and team information exactly.